Statistics – 6

Covariance, Correlation & Probability Distributions

11 April 2026 · Lecture Notes · Statistics-6

Covariance & Correlation	Probability Distributions	PMF / CDF / PDF

Agenda

- Covariance & Correlation
- Probability Distribution
- Demo / worked examples
- Discrete: PMF, CDF
- Continuous: PDF, CDF

§ 01 Covariance

KEY IDEA

Variance / Std Dev tells how spread out *all data in one column/variable* is.

Covariance answers: *"If one variable changes, does another also change?"*

Motivating Example

id	live_class_attended (out of 100)	completion_rate
0	1	≈ 0%
1	100	≈ 100%
2	50	> 50%

Pattern: as **live_class_attended** ↑, **completion_rate** ↑ → directly proportional.

Types of Covariance

Covariance Value	Relationship	Example
Positive (+ve)	Directly proportional ↑↑	live_class ↑ → completion ↑
Zero (0)	No relation	body_weight vs row_index (pandas)
Negative (–ve)	Inversely proportional ↑↓	body_weight ↑ → running_speed ↓

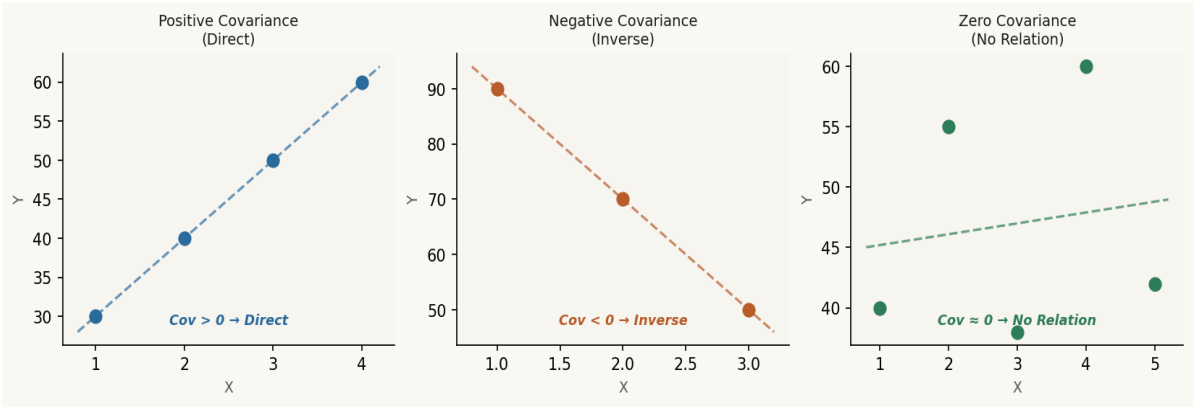

Figure 1: Scatter plots illustrating positive, negative, and zero covariance

Formula

$$\mathrm{Cov}(X, Y) = \Sigma\ (x_{\blacksquare} - x_{\blacksquare})(y_{\blacksquare} - {}_{\blacksquare})\ /\ (n - 1)$$

Worked Example — Study Hours vs Marks

Student	Study Hours (X)	Marks (Y)
A	1	30
B	2	40
C	3	50
D	4	60

```
Step 1: x■ = (1+2+3+4)/4 = 2.5 ■ = (30+40+50+60)/4 = 45
```

Step 2 — Compute deviations and products:

Student	X	Y	(X – x■)	(Y – ■)	Product
A	1	30	1–2.5 = –1.5	30–45 = –15	22.5
B	2	40	2–2.5 = –0.5	40–45 = –5	2.5
C	3	50	3–2.5 = +0.5	50–45 = +5	2.5
D	4	60	4–2.5 = +1.5	60–45 = +15	22.5
				Σ =	50

```
Cov(X,Y) = 50 / (4–1) = 50/3 ≈ 16.67 → Positive → Directly Proportional
```

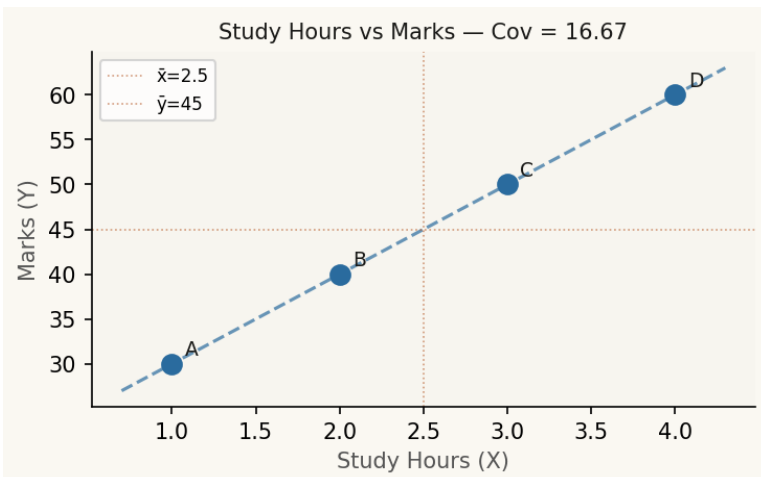

Figure 2: Study Hours vs Marks scatter plot (Cov = 16.67, positive relationship)

Second Example — Hours vs Stress (Negative Covariance)

Hours	Stress	(X – x■)	(Y – ■)	Product
1	90	–1	+20	–20
2	70	0	0	0

3	50	+1	–20	–20
			Σ =	–40

```
–40 / (n–1) = –40/2 = –20 < 0 → Negative covariance → Inversely proportional
```

Problem with Covariance — Units Matter!

ISSUE: UNITS

With 20 columns in ML, covariance between columns with different units is hard to compare.

Example: A–B (hours vs mm) → large value (1000) vs A–C (hours vs marks) → small value (500)

→ Can't say A–B has a 'better relation' just because it's larger — it's a unit problem!

→ Solution: **Normalize** covariance to remove units → **Correlation**

§ 02 Correlation

DEFINITION

Correlation = normalized covariance. It removes units and fixes the range to [–1, 1].

Range: $-1 \le r \le +1$

```
Correlation (r) = Cov(X, Y) / (σ■ · σ■)
σ■ · σ■ → normalizes covariance → removes units
Range: [–1, 1]
```

Worked Example — Correlation Calculation

X	Y	(X – x■)	(Y – ■)	Product

1	30	–1	–10	10
2	40	0	0	0
3	50	+1	+10	10
			Σ =	20

```
Cov = 20 / (3–1) = 20/2 = 10
```

Standard Deviations:

X: deviations² = $(-1)^2,(0)^2,(+1)^2$ = 1,0,1 → Sum=2 → Variance=2/2=1 → **σ■ = √(2/2) = 1**

Y: deviations² = $(-10)^2,(0)^2,(+10)^2$ = 100,0,100 → Sum=200 → Variance=200/2=100 → **σ■ = √(200/2) = 10**

```
Correlation = Cov(X,Y) / (σ■ · σ■) = 10 / (1 · 10) = 10/10 = 1
→ r = 1 → Perfect positive correlation → Directly proportional
```

Correlation (r)	Interpretation
+1	Perfect positive — directly proportional
0.7–0.99	Strong positive correlation
0	No linear relationship
–0.7 to –0.99	Strong negative correlation
–1	Perfect negative — inversely proportional

Practical ML Example (Feature Selection)

FEATURE SELECTION WITH CORRELATION

Dataset with 10 columns (excluding index):

A) ML on all columns → 9 features → **86% accuracy**

B) ML on top 5 correlated columns → **92% accuracy** ← Better!

C) ML on top 3 columns → 91.6% accuracy

→ More features ≠ better. Correlated features + simpler model often wins.

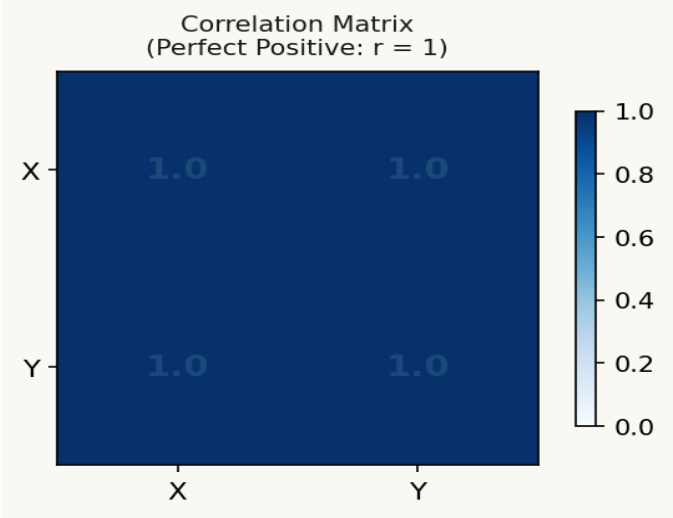

Figure 3: Correlation matrix (all values = 1 → perfect positive correlation)

§ 03 Probability Distributions

OVERVIEW

Discrete RV → PMF (Probability Mass Function) + CDF

Continuous RV → PDF (Probability Density Function) + CDF

PMF — Probability Mass Function

KEY POINT

PMF is a function, not just a plot.

It answers: *"What's the chance of X happening exactly?"*

- Sum of all probabilities = 1
- Only works for **discrete** outcomes

Example — Dice roll: S = {1, 2, 3, 4, 5, 6}

Face	PMF P(X=x)
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

```
def pmf_dice(face):
if face in [1, 2, 3, 4, 5, 6]:
return 1/6
else:
return 0
```

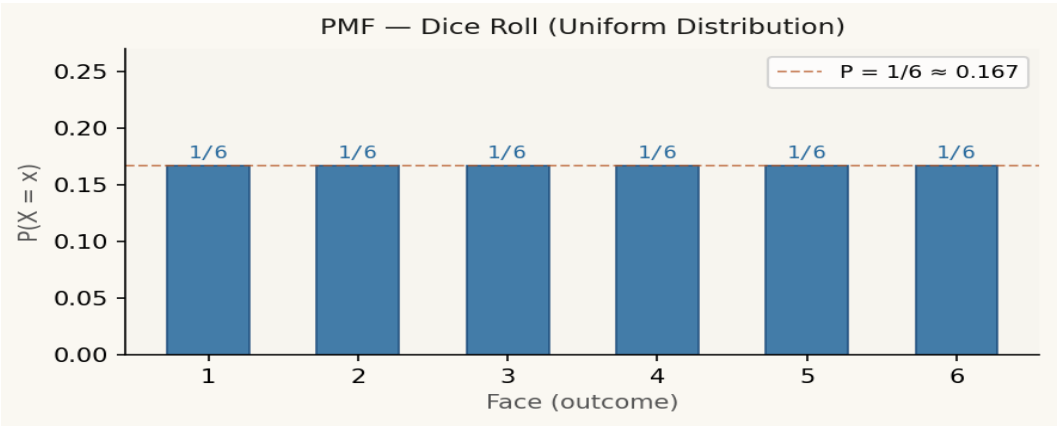

Figure 4: PMF of a fair dice — uniform distribution, P(X=x) = 1/6 for all faces

CDF — Cumulative Distribution Function

DEFINITION

Works for **both discrete and continuous** data.

Gives the probability that X is ≤ **a value**.

Answers: *"What's the chance of X being less than or equal to 5?"*

Example: *"What's the prob of rolling ≤ 5 on a dice?"*

CDF for Dice:

P(X ≤ k)	Calculation	CDF Value
P(X ≤ 0)	0	0
P(X ≤ 1)	PMF(0) + PMF(1) = 0 + 1/6	1/6
P(X ≤ 2)	PMF(0) + PMF(1) + PMF(2)	2/6
P(X ≤ 5)	PMF(1)+…+PMF(5)	5/6
P(X ≤ 6)	PMF(1)+…+PMF(6)	6/6 = 1

CDF PROPERTIES

→ CDF always starts at 0 and ends at 1

→ For discrete data: it's a **step function**

→ For continuous data: it's a **smooth S-curve**

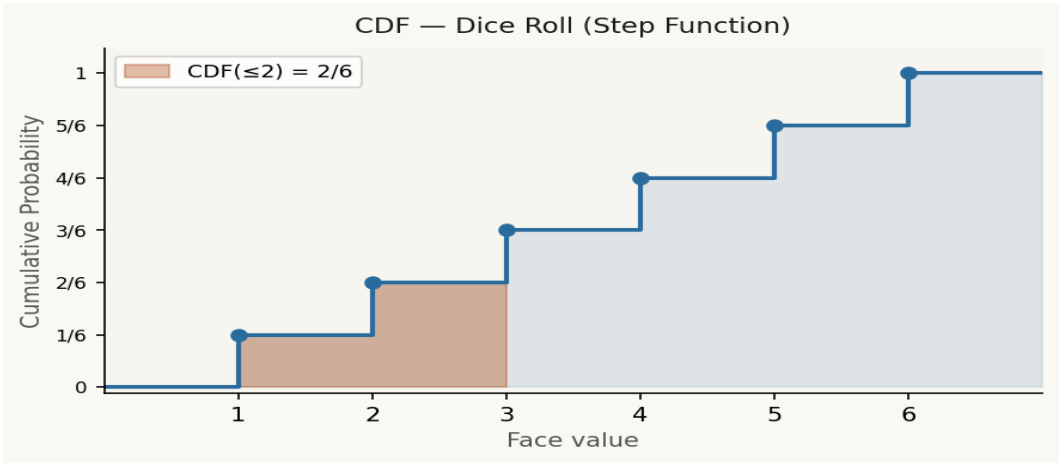

Figure 5: CDF of dice roll — step function, shaded area = CDF(≤2) = 2/6

PDF — Probability Density Function (Continuous)

WHY NOT PMF FOR CONTINUOUS DATA?

For continuous data (height, weight, temperature), there are infinite possible values.

P(height = exactly 160 cm) = 1/∞ = 0 → Exact point probability is meaningless!

→ Instead, ask: *"What's the prob height is BETWEEN 160 cm and 170 cm?"* (a range)

PDF DEFINITION

- PDF is for **continuous** data (height, weight, temperature).
- It shows the **relative likelihood** of a range of values (not exact point).
- **Area under PDF curve = probability** (not the height!)
- Higher PDF value → more concentrated probability around that value.
- Continuous probabilities are only meaningful over **intervals**.
- pdf = density (Not probability)

Building a PDF — Step by Step

Step 1: Collect Data — Heights (cm) of 20 people (min=154, max=179)

```
[162, 168, 171, 154, 175, 178, 165, 169, 172, 160, 167, 173, 170,
176, 164, 179, 166, 170, 174, 168]
```

Step 2: Create a Histogram (Bins × Frequency) → n_bins=6, width=5

Height (cm)	Frequency	PDF = freq / (n × bin_width)
150–155	1	(1/20)/5 = 0.01
155–160	0	0
160–165	3	(3/20)/5 = 0.03

165–170	6	(6/20)/5 = 0.06
170–175	6	(6/20)/5 = 0.06
175–180	4	(4/20)/5 = 0.04

Step 3: Normalize Histogram to get PDF

```
PDF = (frequency of bin) / (total observations × bin width)
= frequency / (20 × 5) = frequency / 100
Example: bin [165–170], freq=6 → PDF = 6/100 = 0.06
```

Step 4: Calculate Area (Probability) under PDF curve

Q: P(165 ≤ height ≤ 170)

→ Identify bin: 165–170 cm

→ Area = PDF value × bin width = 0.06 × 5 = **0.30 (30% chance)**

```
Verification: actual data points in 165–170 = 6 → P = 6/20 = 0.30
✓
```

Example 2: P(height ≤ 170)

Bin	PDF	Area = PDF × 5
150–155	0.01	0.05
155–160	0.00	0.00
160–165	0.03	0.15
165–170	0.06	0.30
Total up to 170		0.05+0+0.15+0.30 = 0.50

```
Verification: people with height ≤ 170 = 10 → P = 10/20 = 0.50 =
50% ✓
```

CDF Formula (Range Query):

$$P(a \le X \le b) = CDF(b) - CDF(a)$$
$$P(165 \le height \le 170) = CDF(170) - CDF(165) = 0.50 - 0.20 = 0.30$$

CDF Cumulative Table — Heights:

Bin	Freq	PDF	Prob (area)	Cumulative CDF
150–155	1	0.01	0.05	0.05
155–160	0	0.00	0.00	0.05
160–165	3	0.03	0.15	0.20
165–170	6	0.06	0.30	0.50
170–175	6	0.06	0.30	0.80
175–180	4	0.04	0.20	1.00

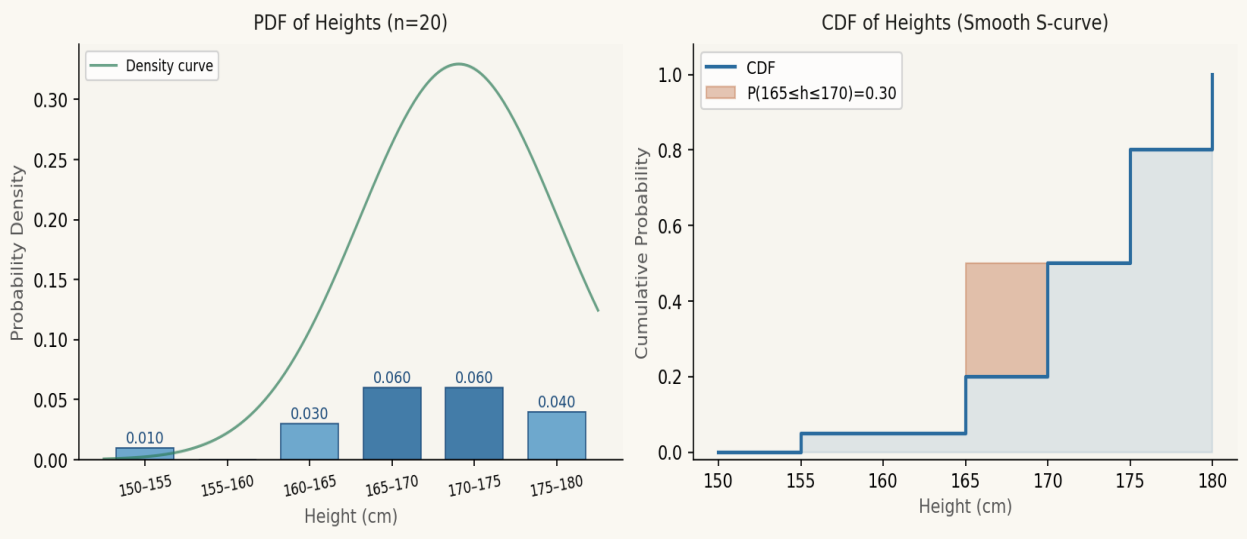

Figure 6: PDF histogram (left) and CDF S-curve (right) for heights of 20 people

§ 04 Summary — Key Formulas & Concepts

Concept	Formula / Rule

Covariance	Cov(X,Y) = Σ(x■–x■)(y■–■) / (n–1)
Cov > 0	Directly proportional (↑↑)
Cov = 0	No linear relationship
Cov < 0	Inversely proportional (↑↓)
Correlation	r = Cov(X,Y) / (σ■ · σ■)
Correlation Range	–1 ≤ r ≤ +1
PMF	P(X=x) for discrete RV; Σ P(x) = 1
CDF (discrete)	P(X≤k) = Σ PMF up to k; step function
PDF	For continuous; area under curve = probability
PDF formula	PDF = frequency / (n × bin_width)
CDF range query	P(a≤X≤b) = CDF(b) – CDF(a)
CDF always	Starts at 0, ends at 1

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